

A Geometric and Bayesian Formulation of Autoregressive Language Models

Decomposing Causal Paths, Latent Stream Variance, and Pre-Training Forces

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Research Presentation

1. The Bayesian Formulation of Language Modeling

Autoregressive models compute the next-token probability $P(w \mid C)$ directly. We invert this via Bayes' Theorem to expose the generative mechanics [2]:

Bayesian Decomposition

$$P(w \mid C) = \frac{P(C \mid w) \cdot P(w)}{P(C)} \propto P(C \mid w) \cdot P(w)$$

Where:

- ▶ $P(w)$ is the **unconditional token prior** (language baseline).
- ▶ $P(C \mid w)$ is the **contextual likelihood**

A Geometric and Bayesian Formulation of LLMs

Generative Terms

Prior $P(w)$

Vocabulary Embedding Norms

Likelihood $P(C \mid w)$

MHA Routing & FFN Memory

2. Likelihood: Multi-Head Attention as Contextual Routing

In layer l , the MHA block computes token-to-token dependencies. For head $g \in \{1, \dots, H\}$ with key-query dimension d_k , the routing weights $\alpha_{n,i}^{(l,g)}$ align the terminal query with preceding keys [1]:

Attention Routing & Likelihood Contribution

$$\alpha_{n,i}^{(l,g)} = \frac{\exp\left(\frac{q_n^{(l,g)} \cdot k_i^{(l,g)}}{\sqrt{d_k}}\right)}{\sum_{j=1}^n \exp\left(\frac{q_n^{(l,g)} \cdot k_j^{(l,g)}}{\sqrt{d_k}}\right)}$$

$$\log P(C \mid w)_{\text{MHA}} \sim \sum_{g=1}^H \sum_{i=1}^n \alpha_{n,i}^{(l,g)} \left(h_i^{(l-1)} W_V^{(l,g)} W_{O,g}^{(l)} u_w \right)$$

This separates dynamic context routing ($\alpha_{n,i}^{(l,g)}$) from semantic value alignment ($h_i W_V W_{O,g} u_w$).

3. Likelihood: FFN as Concept Memory Retrieval

The position-wise Feed-Forward Network (FFN) functions as a key-value associative memory over conceptual rules, mapping context to target tokens [3]:

FFN Associative Memory & Likelihood Contribution

$$\text{FFN}(x_n^{(l)}) = \sum_{m=1}^{d_{\text{ff}}} \sigma \left(x_n^{(l)} \cdot k_m^{(l)} + b_{1,m}^{(l)} \right) v_m^{(l)}$$
$$\log P(C \mid w)_{\text{FFN}} \sim \sum_{m=1}^{d_{\text{ff}}} \sigma \left(x_n^{(l)} \cdot k_m^{(l)} + b_{1,m}^{(l)} \right) \cdot (v_m^{(l)} \cdot u_w)$$

where $\sigma(\cdot)$ measures the scale of **rule satisfaction** for concept m , and $(v_m^{(l)} \cdot u_w)$ represents the **concept-to-token semantic alignment**.

4. Prior: Unembedding Bias and Vector Norms

The unconditional token prior $P(w)$ is stored statically within two parameters during optimization:

1. Unembedding Bias ($b_{U,w}$)

In models featuring explicit unembedding bias, under an uninformative context ($h_n \rightarrow \mathbf{0}$), the prior is:

$$P(w) \approx \frac{\exp(b_{U,w})}{\sum_{v \in V} \exp(b_{U,v})}$$

2. Embedding Vector Norms ($\|u_w\|$)

Decomposing the logit $z_w = h_n^{(L)} \cdot u_w$ geometrically yields [4]:

$$z_w = \|h_n^{(L)}\| \|u_w\| \cos(\theta_w)$$

High baseline frequencies in the corpus drive the vector magnitude $\|u_w\|$ upward, acting as a context-free prior booster.

5. Statistical Mechanics: Latent Stream Variance

The norm of final hidden state $\|h_n^{(L)}\|$ acts as a context-dependent entropy controller:

Activation Variance to Vector Magnitude

If cross-channel mean is centered near zero ($\mu_x \approx 0$) [5]:

$$\|h_n\| = \sqrt{d \cdot \sigma^2}$$

This maps to the physical **inverse temperature** $\beta = \frac{1}{T}$ in Softmax [2, 6]:

$$P(w \mid C) = \frac{\exp(\beta \cdot \|u_w\| \cos(\theta_w))}{\sum_{v \in V} \exp(\beta \cdot \|u_v\| \cos(\theta_v))}, \quad \text{where } \beta = \|h_n\|$$

- ▶ **High Context Certainty:** Constructive interference expands $\|h_n\|$, leading to high β (sharp, low-entropy predictions).
- ▶ **High Ambiguity / Noise:** Destructive interference shrinks $\|h_n\|$, leading to low β (flat, high-entropy predictions).

6. Pre-Training as a Dual-Force Geometric Optimization

Pre-training optimizes cross-entropy loss $\mathcal{L}$ [7].
Gradients generate a dual-force system
organizing the semantic space:

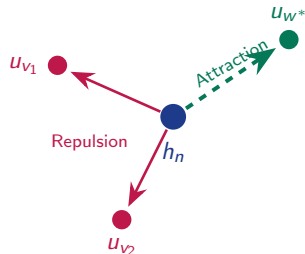
Attraction and Repulsion Forces

$$\frac{\partial \mathcal{L}}{\partial u_{w^*}} = -(1 - P(w^* | C)) h_n$$

Attraction: Rotates context h_n and correct token u_{w^*} toward alignment.

$$\frac{\partial \mathcal{L}}{\partial u_v} = P(v | C) h_n \quad (v \neq w^*)$$

Repulsion: Pushes context h_n away from incorrect token vectors u_v .



7. Conclusion & Distributional Semantics

Distributional Hypothesis Emergence

Semantic substitutions receive identical updates from shared contexts, settling the embedding space into a coordinate system where spatial proximity mirrors semantic overlap [8]:

$$\cos(\theta_{w_1, w_2}) \rightarrow 1 \iff \mathcal{C}_{w_1} \approx \mathcal{C}_{w_2}$$

where $\mathcal{C}_w$ represents the distribution of contexts in which token w appears.

Summary: The Transformer acts as a Bayesian engine where:

- ▶ **Attention & FFNs** synthesize the context vector h_n .
- ▶ The **magnitude** $\|h_n\|$ models certainty as inverse temperature β .
- ▶ The **orientation** matches correlation r with self-organized embeddings.

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